

Project Deliverable C:
Design Criteria and Target Specifications

Group 16: The Sample Snatchers

Ang Li, Eden Irwin, Isaac Pinarski

Radhe Pandey, and Tiam Morrow-Rogers

Faculty of Engineering – University of Ottawa

GNG1103: Engineering Design

Dr. David Knox

February 2, 2025

Table of Contents

1.0 Introduction.....	4
1.1 Introduction.....	4
1.2 Problem Statement.....	4
2.0 Design Criteria and Target Specifications.....	5
2.1 Defining Design Criteria.....	5
2.2 Technical and User Benchmarking.....	5
2.3 Engineering Design Specifications.....	7
3.0 Conclusion.....	8
3.1 General Progress.....	8
3.2 Client Meeting Reflection.....	8
3.3 Trello: Task Plan Update.....	8
Citations.....	9

List of Tables

Table 2.1-1: Design Criteria.....	5
Table 2.2-1: Product Benchmarking.....	6
Table 2.2-2: User Benchmarking - Tasks of CWEST.....	6
Table 2.3-1: EDS - Functional Requirements.....	7
Table 2.3-2: EDS - Constraints.....	7
Table 2.3-3: EDS - Non-Functional Requirements.....	7

Abstract

This report outlines, organizes, and analyzes this project's fundamental design criteria and target specifications. The project aims to develop a compact, fully removable device for extracting metal samples from nuclear reactor pressure tubes.

The document details the prioritized design criteria derived from user needs, incorporating functional and non-functional requirements and key constraints. Technical and user benchmarking have been conducted to investigate and deeply understand the user's potential missed needs and wants for the design. Engineering design specifications (EDS) for the categorized design criteria are established into quantifiable attributes derived from the prioritized needs list. These will be used later to benchmark the prototypes developed.

Conclusively, the report reflects on insights from the preliminary client meeting, and details updates contributing towards the project's task management. The report's content will guide the development and evaluation of ideations, the following stage of the Design Thinking process.

1.0 Introduction

1.1 Introduction

This report centers around the define stage. It aims to create prioritized design criteria, execute technical and user benchmarking, and determine target specifications that will aid in developing our final solution. Previously developed needs will be used to create the necessary criteria for assessing developed solutions for the problem statement.

1.2 Problem Statement

Operators who service nuclear reactors need a device to collect metal samples from pressure tubes while minimizing exposure to contaminated material. The solution should be compact and completely removable from the pressure tube, and it should also be able to report the stage of operation within a limited timeframe.

2.0 Design Criteria and Target Specifications

2.1 Defining Design Criteria

Based on the team's list of interpreted user needs, we have defined the prioritized design criteria as exemplified in [Table 2.1-1](#). This table will be referenced to build the Engineering Design Specifications (EDS) in [Section 2.3](#). The criteria are organized based on the hierarchy of needs done in the previous week.

Table 2.1-1: Design Criteria

Rank	Need	High Priority	Design Criteria
1	Sample is contained to limit exposure	 SAFETY 	Thickness (mm) Material (Density)
2	Design can be completely removed	 SAFETY 	Maneuverable
3	The sample collected can be varied		30mg - 80mg sample collection
4	The design is able to report current stage in it's process		Display feedback
5	Should be modular/lightweight		Weight (lbs) size (cm ³)
6	Necessary power sources should be included		Batteries/Compressed Air/Rat
7	Is a novel creative solution		Unique (by comparison)
8	Device is reasonably fast		Time (s)

2.2 Technical and User Benchmarking

The team performed technical benchmarking by researching existing products that satisfy the interpreted needs, as shown in [Table 2.2-1](#). The group chose to use devices that remove material from narrow passages instead of solely focusing on devices used to collect samples from pressure tubes in reactors. This was due to the innately confidential nature of tooling such as CWEST and the need to try and maintain a creative outlook for the project, as the client strongly suggested creativity as a desirable objective.

From the product benchmarking, a notable feature of the CWEST device is that it is controlled and powered by the BRIMs system, which, for the project, we will not have access to. Including

the Mars Rover in the product benchmarking provides many tangible features to our potential design objective. Other devices were chosen because they meet at least one design criterion.

Table 2.2-1: Product Benchmarking

Device Specifications	Device	CWEST	Arterial Catheter	Drain Snake	CSA420 Endress	Mars Rover - Curiosity
Company		Kinectrics	Generic	Generic	Endress & Hauser	NASA
Cost		\$\$\$\$\$\$\$\$\$	\$	\$	\$\$\$	\$\$\$\$\$\$\$\$\$\$\$\$\$
Stability		BRIMs	Guided by artery	None		Multiple surface contact points
Shape		Tube	Wire	Wire	Tube	Rectangular
Power		External: BRIMs	N/A	External: Man power	External: Man power	Solar + Radioisotope
Failsafe		BRIMs	Pressure Drop	None	None	Safe-mode software, sensors
Controls		BRIMs	Direct hand	Direct Hand	Control panel	Digital signalling
Sampling Method		Scraping	Suction	Friction	Water Pressure	Drill, Scoop
Size		~4 ft	20 gauge	35.5” x 0.45”	381 mm x 75 mm	10’ x 9’7” x 7’
Sample Size		30-80mg	1 mL	N/A	Liquid	45mm³
Material		Zirconium	Polyurethane, Silicone, Nitinol, Stainless Steel	plastic, metal	Polyamid PA, Stainless Steel	7075 Aluminum, Titanium. Aerogel

Task-based user benchmarking was conducted using raw data from the client meeting. This benchmarking helps the team see where each need has its practical situation. The client mentioned overwhelming positivity towards the CWEST device and had no notable changes to note on an empathy map. Additionally, there is insufficient information to create a comprehensive journey map. User benchmarking can also be conducted later in the project to develop a feedback system for our prototypes. This will eventually give a more holistic framework by which we can gauge how users may react to the developed design and act as an effective communication tool for the presentation.

Table 2.2-2: User Benchmarking - Tasks of CWEST

Task	Concern
Enter Reactor vault	Can the device fit in the vault
Attach CWEST device	Weight and maneuverability
Operate CWEST device	Sufficient power, damage to pressure tube
Collect Sample	Exposure to sample
Detach CWEST device	Exposure to contaminated tooling
Analyze Sample	Sample quantity

2.3 Engineering Design Specifications

Design specifications were developed and organized into three categories based on the design criteria highlighted in [Table 2.1-1](#). These categories include functional requirements, constraints, and non-functional requirements. These requirements will be used to evaluate the ideations and prototypes that will be developed.

Table 2.3-1: EDS - Functional Requirements

Design Specifications		Relation (=, < or >)	Value	Units	Verification Method
Functional Requirements					
1	Collect a metal sample	=	30-80	mg	Test
2	Completely Removable	=	Yes	N/A	Test
3	Enclosed Sample	=	Yes	N/A	Test
4	Stage Reporting	<	2	s	Test
5	Visual for process	=	Yes	in, dpi	Test
6	Device speed (time)	<	10	min	Test
7	Power source included	=	Yes	N/A	Analysis
8	Distance Traveled	=	15	ft	Test

Table 2.3-2: EDS - Constraints

Design Specifications		Relation (=, < or >)	Value	Units	Verification Method
Constraints					
1	Weight	<		kg	Analysis
2	Size (Diameter)	<	4	in	Analysis
3	Size (Length)	<	4	ft	Analysis
4	Sample Exposure	<	Yes	N/A	Test
5	Sample Collection	=	30 - 80	mg	Test

Table 2.3-3: EDS - Non-Functional Requirements

Design Specifications		Relation (=, < or >)	Value	Units	Verification Method
Non-Functional Requirements					
1	Aesthetics	=	Yes	N/A	Survey
2	Safety	=	Yes	N/A	Analysis
3	Corrosion Resistant	=	Yes	N/A	Analysis
4	Operation Time	<	10	mins	Test
5	Creative Solution	=	Yes	N/A	Analysis

3.0 Conclusion

3.1 General Progress

Sample Snatchers has established a structured list of design criteria, prioritizing safety, removability, and maneuverability. The engineering design specifications tables include at least one design criterion for each identified need. Benchmarking had no immediate significant impact on the previously completed Deliverable B but does give a frame of reference for future concepts. From benchmarking, the components required for a functioning device can be determined. It has also allowed different sampling methods and their implementation to be considered. The group has also made progress in developing ideations for subsystems for use in Deliverable D. Future work will be able to include a validation process for developed designs by using the content of this report.

3.2 Client Meeting Reflection

As the team developed the design criteria, the initial client meeting provided useful communicational indications to guide the ranking of these criteria' relative importance and create a problem statement. In Deliverable B, the following sequential process was performed: collecting and interpreting raw data from the client meeting, formulating critical user statements, transforming user statements into interpreted user need statements, and establishing importance. Based on the hierarchy of the interpreted needs, the developed design criteria provide descriptive guidance of what the product must be.

3.3 Trello: Task Plan Update

In Trello, the team's task board has been updated to include changes in task duration, missing tasks, task responsibilities, and further details to maintain effective project management. The group has begun on Ideations to develop concepts for Deliverable D.

Please refer to the following link to review updates:

<https://trello.com/invite/b/67846e4e793783740d87aed9/ATTIbb8224a0c532547c0a69acd41093c7c7A4D60EC7/gng1103-project-sample-snatchers>

Citations

Amazon.com. (2025, February 1). *Customer reviews for 35.5 inch Drain Clog Remover*.

Retrieved February 1, 2025 from

https://www.amazon.com/product-reviews/B0C3L7FKCQ/ref=cm_cr_ar_p_d_viewopt_kywd?filterByStar=five_star&reviewerType=all_reviews&pageNumber=1

Cotterill, P. (n.d.). *The hydrogen embrittlement of metals*. Progress in Materials Science, 9(4),

205-250. [https://doi.org/10.1016/0079-6425\(61\)90005-6](https://doi.org/10.1016/0079-6425(61)90005-6)

KINECTRICS (2021, January 18) Fuel Channel Inspection Tools. Youtube

<https://www.youtube.com/watch?v=J-Pnytp9b2g&t=0s>

Custom circumferential wet scrape tool (CWEST) Design & Supply. Kinectrics. (n.d.).

<https://www.kinectrics.com/projects/updated-design-supply-of-a-custom-circumferential-wet-scrape-tool>

Fuel Channel Inspection & Maintenance. Kinectrics. (n.d.-b).

<https://www.kinectrics.com/capabilities/products/nuclear-generation-products/fuel-channel-inspection-maintenance>

Assessment and management of ageing of major nuclear power plant components important to safety: Candu Reactor Assemblies. (n.d.).

https://www-pub.iaea.org/MTCD/Publications/PDF/te_1197_prn.pdf